



सत्यमेव जयते
NITI Aayog



SECTOR AT A GLANCE 06

Blue Economy in India

Water in India

DRAFT

SECTOR AT A GLANCE 06

BLUE ECONOMY IN INDIA

WATER IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Blue economy in India. NITI Aayog Sector-at-a-Glance Series – Water in India, 2023 - Water-SAG-06, <link>

Copyright © 2023, NITI Aayog, Government of India

All rights reserved. This draft document and its contents are internal to NITI Aayog. They cannot be reproduced in any manner either in full or parts, thereof. The draft document is subject to further refinements.

All information, ideas, views, opinions, estimates, suggestions, and recommendations (hereinafter 'content') in this draft "portal resource" should neither be understood as professional advice in any manner nor interpreted as policies, objectives, opinions, or suggestions of NITI Aayog. Readers are advised to use their discretion and seek professional advice before taking any action or decision based on these contents. The contents in this portal have been obtained or derived from reliable sources as far as possible. But, NITI Aayog neither represents this information to be accurate nor assumes any liability for any loss, or damages, caused due to any reason whatsoever, towards its usage by a third party. Readers of this document are made aware that the document may be subject to revisions. Any suggestions/input may please be sent to nitiforstates@gov.in.

www.niti.gov.in

Overview

The blue economy refers to the sustainable use of ocean, sea and coastal resources for economic growth, improved livelihoods, and ecosystem health.¹ The concept emerged around the 2010s as oceans were recognised as drivers of economic growth and vital to achieving UN Sustainable Development Goals.² When sustainably developed, oceans can support food security, jobs, energy access, materials and product access via shipping and mining, and economic opportunities – key priorities for India and countries worldwide. According to the OECD, the blue economy could contribute USD 3+ trillion to the global economy by 2030.³

India has started its journey of harnessing the ocean economy. However, certain exploitative practices such as overfishing and pollution, conflicts between different sectors over ocean space and resources, limited financial resources, and the climate impacts on oceans need to be addressed.

Given these opportunities and challenges, establishment of port infrastructure, development of deep-sea technologies, conservation of ocean biodiversity and harnessing offshore wind energy are major policy focus areas. Several ongoing and recent central government initiatives including but not limited to, the Deep Ocean Mission and the SagarMala program focus on harnessing India's blue economy. These efforts, along with several state and local-level policy initiatives seek to ensure that India's blue economy can drive substantial growth over the coming decades.

This note presents key statistics, challenges, opportunities, and some ideas for the way forward that could help India realize the full potential of its Blue Economy.

Figure 1: Key blue economy statistics

7,500 kms

Coastline length of India

30%

Coastal states in India

4 MN+

Fisherfolk and coastal communities sustained by the coastal economy

95%

India's trade volume transiting by sea

4% of GDP

Estimated size of India's blue economy

Source: Economic Advisory Council to the Prime Minister, India's Blue Economy: Draft Policy Framework, 2020

1 UN, Blue Economy Definitions, accessed July 2023

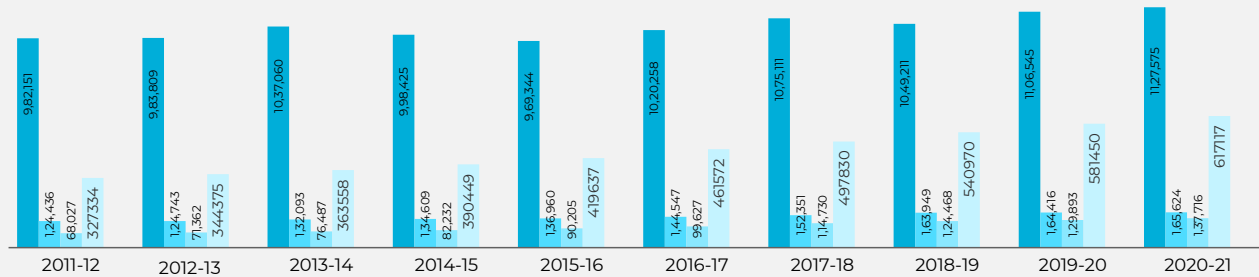
2 UN, Blue Economy: Oceans as the Next Great Economic Frontier, 2022

3 OECD, The Ocean Economy in 2030, 2016

Figure 2: Gross value added (in INR crore) by the fisheries economy at constant (2011-12) prices (2011-12—2020-21)

Constant Prices (2011-12)

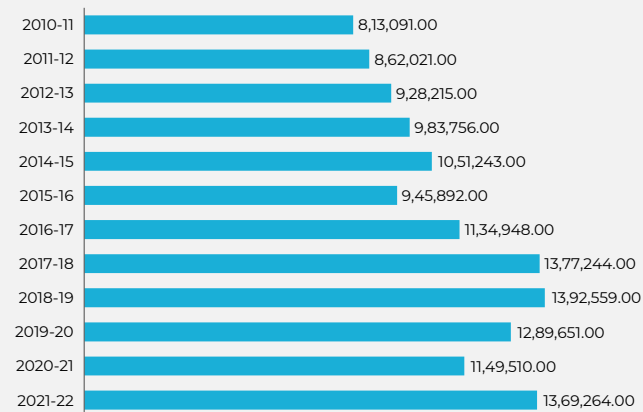
Crops Forestry and Logging Fisheries Livestock



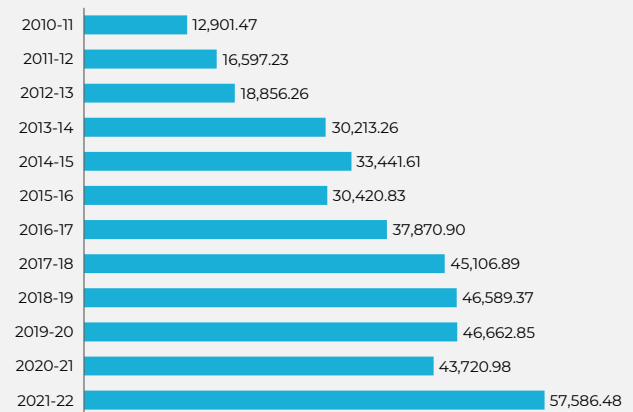
Source: Department of Fisheries, Handbook on Fisheries Statistics 2022, 2023

Figure 3: Export of fish and fish products in quantity (tonnes) and value (INR crore) (2010-11—2021-22)

Quantity (tonnes)

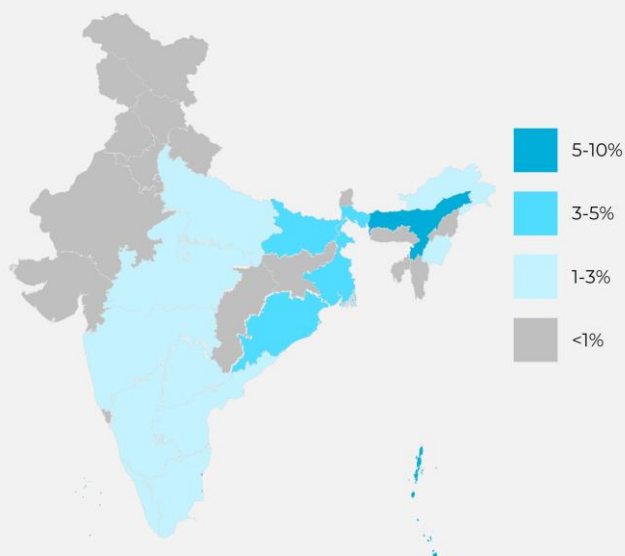


Value (INR crore)



Source: Department of Fisheries, Handbook on Fisheries Statistics 2022, 2023

Figure 4: Percentage share of fisherman population by state and UT (2022)



Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by promoting India's blue economy. These revolve primarily around the potential for fisheries and aquaculture, ports and shipping, ocean energy and tourism.



Growth of fisheries and aquaculture sectors:

Fisheries and aquaculture employ over 14 million people in India and have tremendous potential for growth.⁴ The Pradhan Mantri Matsya Sampada Yojana was launched by GOI in May 2020 with an outlay of more than INR 20,000 crores to grow the fisheries sector in a sustainable, responsible, and equitably.⁵



Expansion of ports and shipping infrastructure:

Over 90% of India's trade by volume passes through maritime transit.⁶ Investing in ports, coastal economic zones, domestic shipping, and shipbuilding can strengthen logistics and trade.



Harnessing ocean energy:

Offshore wind, wave, thermal, solar, and tidal energy have significant energy generation potential in India, with the offshore wind itself having 54 GW potential, but technology and infrastructure require significant investment to harness this.⁷



Expanding ocean-linked tourism:

Globally, coastal, and marine tourism accounts for at least 50% of global tourism.⁸ India is poised for fast, sustainable coastal and marine tourism growth, especially around islands.⁹ Promoting eco-tourism, beach activities, and cruise tourism have been identified as priorities, alongside mining, biotechnology, and undersea cabling.¹⁰



Enhancing international cooperation:

Regional cooperation between Indian Ocean countries can enhance data sharing, collaborative research, environmental protection, and disaster risk management associated with oceans.

Challenges

This section highlights some of the major challenges and constraints facing the blue economy in India. Efforts to resolve such challenges will ensure optimum utilisation of natural resources, greater internal collaboration, and higher investment.



Exploitative practices affecting natural resources:

4 National Fisheries Development Board (NFDB), Fish and Fisheries of India, 2020; Invest India, India's Blue Economy, 2022

5 Ministry of Fisheries, Animal Husbandry & Dairying, Pradhan Mantri Matsya Sampada Yojana, 2020

6 Ministry of Shipping, Annual Report, 2019-20, 2020

7 Mongabay, India's tidal power potential hampered by high costs and environmental risks, 2021

8 The Ocean Panel, Opportunities for Transforming Coastal and Marine Tourism: Towards Sustainability, Regeneration and Resilience, 2020

9 Economic Advisory Council to the Prime Minister, India's Blue Economy: Draft Policy Framework, 2020

10 Economic Advisory Council to the Prime Minister, India's Blue Economy: Draft Policy Framework, 2020

Overfishing, pollution, coastal erosion, and habitat destruction threaten fish stocks and the health of coastal and marine ecosystems. 36.3% of fish stock in India is overfished, while another 26.5% are recovering, and 3.1% are in the overfishing status.¹¹



Conflicts between sectors connected with the blue economy:

Ocean space and resources are limited, and sectors like fisheries, aquaculture, coastal tourism, shipping, and offshore energy risk being in conflict.¹²



Climate change-related threats to blue economy:

Climate change affects oceans and seas in several ways, such as an increase in ocean temperatures, acidification, changes in fish population distribution, rise in sea levels, and increase in the prevalence of extreme weather events, posing a serious threat to food, economic and human security of the country's coastal population.¹³



Limited public financial resources:

Greater private sector involvement through public-private partnerships (PPPs) and private investments in emerging industries like ocean energy, tourism, mining (including petrochemical extraction), ports, and shipping is needed, given government budget constraints.¹⁴

¹¹ ICAR - Central Marine Fisheries Research Institute, Marine Fish Landings in India – 2021, 2022

¹² FICCI, TERI, Blue Economy: India's Pathway to Sustainable, Secure, and Resilient Economy, accessed July 2023; ORF, Blue Economy in the Indian Ocean: Governance perspectives for sustainable development in the region, 2019

¹³ FICCI, TERI, Blue Economy: India's Pathway to Sustainable, Secure, and Resilient Economy, accessed July 2023; ORF, Blue Economy in the Indian Ocean: Governance perspectives for sustainable development in the region, 2019

¹⁴ Invest India, India's Blue Economy, 2022

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee India's blue economy.



Ministry of Ports, Shipping and Waterways

The ministry focuses on enhancing the maritime infrastructure, promoting port-led development, and facilitating trade through efficient shipping and navigational channels. It works on developing coastal infrastructure, optimizing port operations, and ensuring sustainable utilisation of waterways for economic activities like fisheries and aquaculture.



Ministry of Environment, Forests and Climate Change

The ministry oversees the conservation and sustainable management of marine resources, coastal ecosystems, and biodiversity. It aims to protect marine habitats, regulate coastal development to prevent degradation, and formulate policies for sustainable use of marine resources, aligning with the principles of the blue economy.



Ministry of New and Renewable Energy

The ministry promotes renewable energy sources like offshore wind energy and ocean thermal energy conversion (OTEC). It explores and develops renewable energy technologies that harness the potential of the oceans and seas, contributing to both energy production and sustainable utilisation of marine resources.



Ministry of Earth Sciences

The ministry focuses on scientific research and exploration related to oceans, including oceanography, marine geology, and climate studies. It studies marine ecosystems, predicts weather patterns, and monitors sea-level changes, which are vital aspects of managing the blue economy sustainably.

Major policies and programs

This section highlights existing policies related to the blue economy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Sagarmala Program¹⁵

The program is run by the Ministry of Ports, Shipping and Waterways. It is the flagship program of the government of India that aims to develop new ports and coastal economic zones to promote port-led development.¹⁶ The goal is to drive port capacity to hand 3300+ Million Metric Tonnes per Annum by 2025.¹⁷

¹⁵ Ministry of Ports, Shipping and Waterways, About Sagarmala, Accessed December 2023

¹⁶ Ministry of Ports, Shipping and Waterways, Port Modernization & New Port Development, accessed July 2023

¹⁷ Ministry of Ports, Shipping and Waterways, Port Modernization & New Port Development, accessed July 2023

Deep Ocean Mission¹⁸

Under the Ministry of Earth Sciences, this mission was approved in 2021 with a budget of INR 4,077 crores over five years.¹⁹ It seeks to develop technologies like manned submersibles and deep-sea mining to explore seabed resources, promote deep ocean biodiversity, develop ocean climate change advisory services, and other benefits.²⁰



Conservation and Management of Mangroves and Coral Reefs, National Coastal Mission Programme²¹

Under this mission, an annual Management Action Plan (MAP) for the conservation as well as management of mangroves are to be formulated and implemented across all coastal States and Union Territories in the country.



Island Coastal Regulation Zone (ICRZ) Notification, 2019

This notification declares designated areas as ICRZs, with the aim of conserving and protecting the environment of marine areas and coastal stretches, alongside the livelihood of local communities in such areas, to enable sustainable development accounting for the effects of global warming.²²



Offshore wind energy harnessing programs

The Government has proactively sought to increase offshore wind energy production in India, and its efforts are being led by the Ministry of New and Renewable Energy. A potential of 35 Gigawatts (GWs) and 36 GW offshore production has been identified off the coasts of Tamil Nadu and Gujarat, respectively.²³ The country has set a target of installing, by 2030, 30 GW of offshore wind projects.²⁴

¹⁸ Ministry of Earth Sciences, DOM, Accessed December 2023

¹⁹ Cabinet Committee on Economic Affairs (CCEA), Cabinet approves Deep Ocean Mission, 2021

²⁰ Cabinet Committee on Economic Affairs (CCEA), Cabinet approves Deep Ocean Mission, 2021

²¹ Ministry of Environment, Forest & Climate Change, Conservation and protection of mangroves, 2022

²² Government of Andaman & Nicobar, Preparation of Island Coastal Regulation Zone (ICRZ) Plan, accessed August 2023

²³ Ministry of New and Renewable Energy, Offshore Wind, accessed 24 July 2023

²⁴ Ministry of New and Renewable Energy, Offshore Wind Energy in India, 2019

Way forward

India has made a significant commitment to realizing its blue economy potential through the measures and institutional structures outlined in this note. However, these are still early stages in its journey, as is the case for many other countries.

Some ideas that could accelerate progress going forward include:

 Tapping into blue economy financing mechanisms	Indian states can take inspiration from a range of sustainable blue economy financing principles that have been developed by organizations such as the United Nations Environment Programme.²⁵ This would enable optimal achievement of blue economy goals while at the same time protecting and conserving marine ecosystems and also ensure equitable benefits to a wide range of stakeholders in India.
 Setting up cross-sectoral convenings	Establishing cross-cutting convenings and councils with participation across a range of sectors – tourism, ports, fisheries, civil society, researchers – can help develop blue economy-related plans at the state and sub-state level. Ensuring that a range of perspectives are included while developing policies and programs will be vital given the nature of marine ecosystems and resources.

²⁵ United Nations Environment Programme, Sustainable Blue Finance, accessed December 2023

Recommended readings

Konrad Adenauer Stiftung, The Energy and Resources Institute (TERI), National Maritime Foundation (NMF) and Federation of Indian Chambers of Commerce and Industry (FICCI). (2022). *Blue Economy: India's Pathway to Sustainable, Secure, and Resilient Economy*.

https://www.teriin.org/sites/default/files/2022-07/Blue_Economy_Publication.pdf

United Nations. (2022). *Blue Economy: oceans as the next great economic frontier*.

<https://unric.org/en/blue-economy-oceans-as-the-next-great-economic-frontier/#:~:text=The%20UN%20first%20introduced%20%E2%80%9Cblue,productive%20when%20they%20are%20healthy>

Economic Advisory Council to the Prime Minister. (2020). *India's Blue Economy: Draft Policy Framework*.

https://incois.gov.in/documents/Blue_Economy_policy.pdf



सत्यमेव जयते

NITI Aayog

www.niti.gov.in